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Patient: Zherlitsyna Anastasia Stanislavivna
Date of birth: 25.04.1976
Sex: Female

ORDER: 22B104
Received: 15.12.2022
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Service: OncoMine - NGS study identifying thousands of gene hotspots and variants (SNVs, CNVs, InDels, TMBs, MSIs, HRDs, gene fusions) associated with response to targeted and immunotherapy

Block number: 48278/22
Number of tumor cells: 500-700
Percentage of tumor cells: ~80%

Results of molecular profiling

Identified gene variants (mutations, translocations, amplifications)

Gene	The detected variant	Allele frequency (%) / Number of copies	Biological significance	Therapeutic significance			
				Yes/No	If "yes"		
					OncoKB score	A group of substances	Sensitive./ Resist.
MTAT-CDKN2B_AS1_004	Translocation MTAT-CDKN2B_AS1_004.M5A5	70358 lines per 1 million	Active	No	-	-	-
KIF5B-MET	Translocation KIF5B-MET.K24M15	263 lines per 1 million	Active	Yes	4	MET Inhibitors	Sensitive
BRD3-NUTM1	Translocation BRD3-NUTM1.B10N2	284 lines per 1 million	Active	No	-	-	-
KIT	Mutation c.1668_1676delGTGGAAGGT (p.Gln556_Val559delinsHis)	42,49%	Active	Yes	3B	KIT Inhibitors	Sensitive
TPMT	Mutation c.719A>G (p.Tyr240Cys)	39,83%	Active	No	-	-	-
TPMT	Mutation c.460G>A (p.Ala154Thr)	49,93%	Active	No	-	-	-
PPM1D	Mutation c.1080G>C (p.Gln360His)						

Genomic indicators (TMB, MSI, LOH)

Genomic marker	Result	Therapeutic significance
Mutational load (TMB)	Low (4,82 Mut/Mb)	Low sensitivity to inhibitors PD-1/PD-L1
Microsatellite instability (MSI)	The microsatellite tumor is stable (MSS)	Low sensitivity to inhibitors PD-1/PD-L1
Loss of heterozygosity (LOH)	Unspecified	-

Summary of molecular profiling results

A mutation in the 11th exon of the KIT gene was detected in the tumor. KIT inhibitors such as imatinib, sunitinib, regorafenib, and ponatinib have been approved for patients with gastrointestinal stromal tumors (GIST) with such mutations. The clinical efficacy of these drugs in patients with leiomyosarcoma is currently unknown, but from a biological point of view, they may be potentially effective in this case.

Translocation of the MET gene was also detected in the patient. This translocation leads to the activation of the met protein, so such a tumor can potentially be sensitive to the use of MET inhibitors. However, given the low number of RNA with this translocation (263 lines per 1 million), it can be assumed that this variant is subclonal, which makes it less likely to obtain a clinical effect from targeted therapy. Furthermore, MET gene translocations are currently not a registered indication for targeted therapy in any type of cancer.

In addition, mutations in the TPMT, PPM1D genes and MTAT-CDKN2B_AS1_004 and BRD3-NUTM1 translocations were found in the tumor, which currently have no clinical and therapeutic significance.

When assessing genomic parameters, the tumor was found to be microsatellite-stable (MSS) and to have a low mutational load (TMB-Low), which is associated with a low potential efficacy of immunotherapy with PD-1/PD-L1 inhibitors. The level of loss of heterozygosity (LOH) could not be determined due to insufficient DNA quality in the sample, but given the absence of mutation in the BRCA1/2 genes and other genes of the homologous recombination (HRR) system, the tumor is potentially insensitive to PARP inhibitors.

*Method: Next Generation Sequencing (NGS)
 *Set: Oncomine Comprehensive Assay Plus (Thermo Scientific)
 *Sensitivity: 5% of mutant alleles
 *Equipment: The Ion Proton Sequencer (Thermo Scientific)

Researched gene biomarkers	Researched genomic biomarkers
165 genes with mutations (SNVs, In/Dels) in "hot spots"	HRD – Determination of LOH (loss of heterozygosity) — at the gene level and sample level
333 genes for amplification and deletion (CNV)	>1 mb of coding sequences to determine TMB (tumor mutational load)
227 genes with complete coding DNA sequence (CDS)	MSI-H/MSS microsatellite markers to determine MSI status
49 genes for translocation (Fusions)	
MET detection of exon skipping at the level of DNA and RNA	

*Quality control: The coverage quality of target sequences is satisfactory
*Classification of biological significance: Oncomine database (Thermo Scientific)
*Classification of therapeutic significance: OncoKB (Memorial Sloan Kettering Cancer Center)

*** Research results are not a diagnosis and require a doctor's consultation.**

Performers:

Geneticist Shapochka D.O

